

THE NORTHERN RHODESIAN COPPERBELT:
IS IT A CLASSIC EXAMPLE OF SYN-
GENETIC DEPOSITION¹

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At the Symposium on "Syngenesis versus Epigenesis" in New York in February, 1962, Dr. J. J. Brummer spoke on the syngenetic theory of ore genesis as applied to deposits in Northern Rhodesia. Many of Dr. Brummer's remarks were based upon the volume, "Geology of the Northern Rhodesian Copper Belt," edited by Dr. F. Mendelsohn, Chief Geologist of the Roan Antelope Copper Mines, Ltd., and published in 1961 by MacDonald and Co. of London in time for the Seventh Mining and Metallurgical Congress.

Dr. Brummer unquestionably has a wide knowledge of the district. His thesis in 1955 described the "Geology of the Roan Antelope Mine" and was presented at a meeting of the Institution of Mining and Metallurgy (1). He is also the author of several unpublished company reports. He contributed four photomicrographs of polished sections of ore from Roan Antelope, which, incidentally, are the only ones in the volume.

Since the meeting the writer has critically reviewed the volume on the "Geology of the Northern Rhodesian Copper Belt." The following comments are offered to counter the impression left at the meeting by the proponents of syngenetic theory that:

1. The volume "Geology of the Northern Rhodesian Copperbelt" can be used as the "bible" of that creed called syngeneticism, a belief accepted today by most South African and Australian geologists with a fervor akin to that inspired by religious rather than lay dogma.
2. That all of the authors who contributed sections of the volume, believed in that creed without question.
3. That there was ample supply of copper and cobalt in solution in the ocean waters at the time the enclosing sediments were deposited to precipitate all of these metals, which came from the erosion of a nearby land mass.
4. That there is no question in anyone's mind but that the mineral deposition preceded the intense folding of the Precambrian formations.
5. That recrystallization accompanying this folding involved no introduction of new material, but was simply a rearrangement of that already present.
6. There was no mention of the fact that similar copper deposits are found in Katanga across the border from Northern Rhodesia, and that these deposits occur in beds of a higher stratigraphic horizon, and different lithological character than those in Northern Rhodesia.

¹ This paper originated as a Discussion at the Symposium on "Syngenesis versus Epigenesis" held in New York, Feb. 14, 1962, under the joint auspices of the Society of Economic Geologists and the Society of Mining Engineers of the American Institute of Mining, Metallurgical and Petroleum Engineers.

May I take the liberty of discussing these six points, not with the hope of changing any views already held, but to point out to a rising generation of students of economic geology that they should read the whole book, and not part of it, and other writings that have been omitted in the preparation of "The Geology of the Copperbelt."

Point 1.—This is the thesis that the volume is the "bible" of the creed of syngeneticism.

The principal disciple of that creed is Dr. W. G. Garlick, who describes the theory on pages 145 to 165 of the volume. The editor, Mr. Mendelsohn, however, in introducing the section by Dr. Garlick, warned the reader that "there are some features that a syngenetic theory does not adequately explain."

In developing his hypothesis, Dr. Garlick starts with the analogy of the Kupferschiefer of Mansfield, Germany, which is cited by him as the one incontrovertible example of a sedimentary sulfide deposit of commercial importance. Ignoring the blasphemy that there are still a few agnostics (2) who do not accept this origin of the Kupferschiefer as so obvious a fact, nevertheless Dr. Garlick is not careful to caution his readers that most of the ore beds of the Copperbelt are not black shales like the Kupferschiefer, but rather arenaceous beds with much cross bedding and some are dolomites. Moreover a deposit of sedimentary sulfides should, if the Kupferschiefer is the type example, be very widespread, of uniform composition, should have pyrite as the principal sulfide, and in addition to copper should carry lead and zinc, and should have as much nickel as cobalt (3). These are not the features of the Copperbelt.

Dr. Garlick explains that he was driven to the thesis of syngenetic origin of these deposits, since no igneous rock is exposed and no channelways or fissures through which solutions came, have been found. He emphasizes, of course, the very remarkable confinements of the ore to one sedimentary formation (Fig. 8), although he must bring out sooner or later that the ore occurs up and down through that formation, even in the same mine. Furthermore, barren zones have exactly the same lithology, texture and mineralogy as some of the richest zones.

Dr. Garlick ignores the gabbro, which is found near each major deposit, and seems oblivious to the fact that large masses of gabbro are, in many parts of the world, an offshoot from a large batholith in depth. He is also oblivious to the fact so commonly observed around large igneous masses, that large scale transfer of material, seemed to require no visible channelways.

In developing his hypothesis, Dr. Garlick had much trouble explaining the wide spreading zoning. He said that it is a function of facies changes, and cites examples at Baluba, Nkana and Chambishi.

In describing Nkana, later in the volume, J. Jordaan said that "at Mindola, the mineral zoning is not associated with changes in facies in the sediments . . . but it is in the North and South orebody."

That the zoning problem is complicated is indicated by the description of Nkana in "which cobalt is concentrated in the chalcopyrite zone at Mindola, but at the North orebody, the highest concentrations of cobalt, are in the mixed bornite-chalcopyrite zone. At places, the zonal arrangement is pyrite, then

chalcopyrite and carrollite, followed by mixed bornite, chalcopyrite and carrollite, and finally chalcopyrite and carrollite at the base."

Dr. Garlick explains his idea of zoning on page 155 of the volume, but although this may be an idealized plan of the zoning, it does not prove a case for syngeneticism, in the opinion of this critic.

Dr. Garlick then attempts to flatten out the tightly folded sedimentary members of the Roan series, and show the relation of the present ore zones to what may have been their original position as nearly flat lying beds on a sea bottom close to an old shore line. For formations as tightly folded as these are now, this unrolling back to Precambrian times takes some doing. The

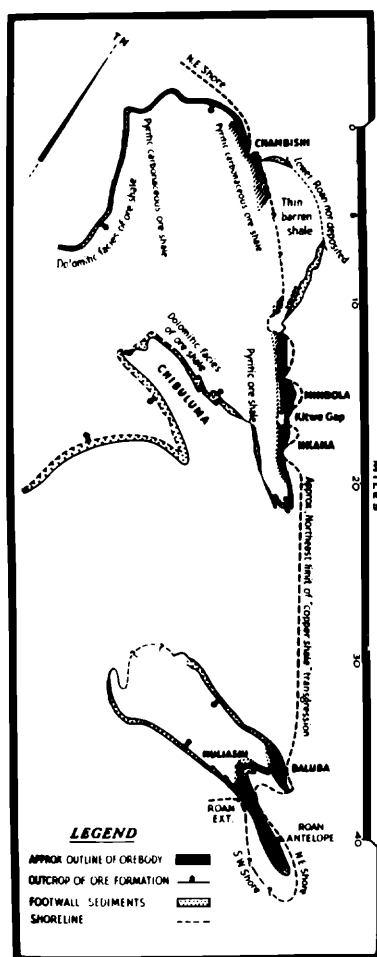


FIG. 1. Ore deposits and pyritic and barren zones related to the shore line from Chamibishi to Roan Antelope. Reproduced from Figure 48, page 157, in the volume "The Geology of the Northern Rhodesian Copper Belt," in section on the Syngenetic Theory, written by W. G. Garlick.

map on page 157 shows only the area on the west side of the Kafue anticline from Roan Antelope to Chambishi. Whether it is typical of the many scores of other miles on the two side of the anticline is inferred, but not proven. Figure 48 is reproduced here as Figure 1.

The final clincher of the syngenetic theory is that the present textures of the ores are the result of recrystallization under heat and pressure of metamorphism, during which there was absolutely no introduction of material.

This will be discussed under Point 5.

Point 2.—This is the tenet that each of the contributors to sections of the *Geology of the Northern Rhodesian Copperbelt* subscribe to the Syngenetic hypothesis. Let us examine this statement.

B. B. Brock, author of part of Chapter 4, "The structural setting of the Copperbelt," definitely disagrees, and most forcefully. On page 89 he wrote:

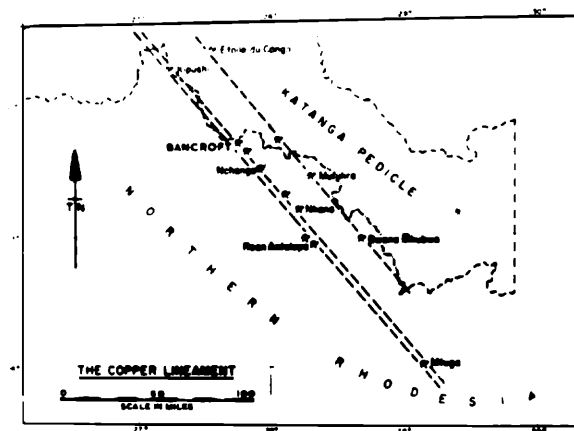


FIG. 2. The main copper lineament through Mtuga, Roan Antelope, Nkana, Nchanga, Bancroft, and Kipushi; and the subsidiary lineament through Bwana Mkubwa, Mufulira, and Etoile du Congo. Reproduced from Figure 26, page 86, in the volume "The Geology of the Northern Rhodesian Copper Belt," in section on the Structural Setting of the Copper Belt, written by B. B. Brock.

The perfection of the copper lineament² over a distance of 200 miles or more, must be nearly unique in the world. . . .

Any theory regarding the origin of the copper that ignores this remarkable alignment might be laying itself open to a charge of arbitrary selectivity of data. . . .

The metallogenic province with which we are dealing, of which the Northern Rhodesian Copperbelt forms only a part, when viewed in the overall view of Africa, is very much of a unity . . . it must be rare in the history of economic geology, that such an important mineral province has never been comprehensively treated as an entity.

Brock's diagrams on the structural framework of Northern Rhodesia, and on the main copper lineament are reproduced here as Figure 2 and on the geologic map, Figure 7 and the structural setting of the area is shown in Figures 3 and 4.

² American readers may understand the word alignment better than lineament, with due apologies to Dr. Brock, and the English language.

F. Mendelsohn, Chapter 7 on Ore Genesis and Chapter 11, on Roan Antelope; he believes in syngensis.

W. G. Garlick, Parts of Chapters 7 and 8 and Chapter 9 on Chambishi; Dr. Garlick of course believes in syngensis.

J. E. G. Schwellnus, Chapter 9, Bancroft-Nchanga.

One comment of his is pertinent:

in the northern part of the mine, three bore holes show poor mineralization in the undeformed ore units, and pronounced chalcopyrite-bornite mineralization along the slaty cleavage directions both within and across the bedding in the more highly deformed units. Although this does not suggest an epigenetic source for the ore, it does indicate a redistribution of primary ore minerals by dynamic metamorphism.

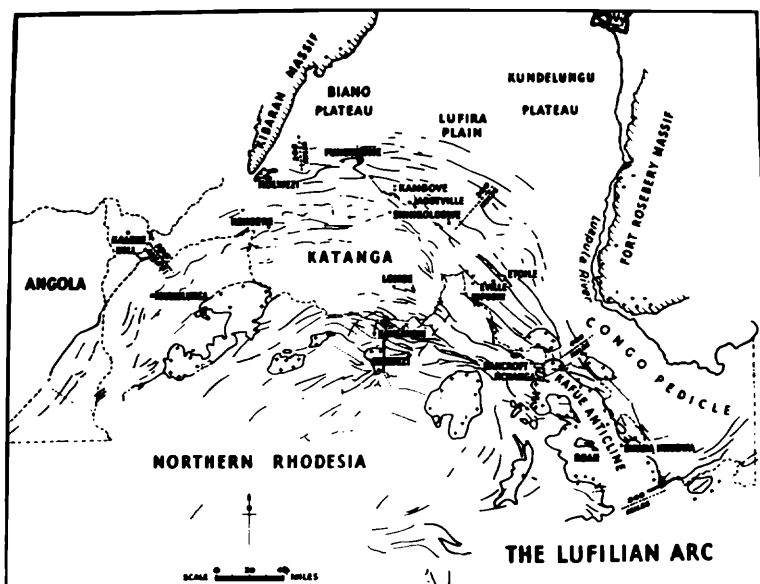


FIG. 3. The Lufilian Arc. Reproduced from Figure 27, page 90, of the volume "The Geology of the Northern Rhodesian Copper Belt" in the section on Structural Evolution, written by W. G. Garlick. This shows the position of the Kafue Anticline.

D. M. McKinnon and N. J. Smit, Part of Chapter 9, Nchanga. A number of the descriptions and comments of these gentlemen are so pertinent, that longer quotations are made from their writings.

The host rocks (page 256) include a wide range of sediments of continental and marine origin, and the deposits are all found in more or less folded structures. . . . At Nchanga, ore can be traced almost continuously in places from granite through more than 300 feet of sediments, including quartzite, sandstone and quartzite, boulder conglomerate, arkose, Lower Banded shale, micaceous Banded sandstone, Feldspathic quartzite and Upper Banded Shale . . . zoning can be seen across the strata.

Ore genesis (page 267) The base metal province in which the deposit occurs must be studied . . . and not piecemeal like a blind man describing an elephant only after feeling his tail. An arc-shaped belt of deformed rocks extends more than 400 miles through Rhodesia and Katanga and it is within this arc that the copper belt rocks occur . . . but the belt is itself contained within a base metal province stretching to the southwest, which covers more than 80,000 square miles, and contains at least 300 known occurrences of copper minerals . . . and each occurrence can be related to a bit of exposed granitic and allied rocks which extends . . . through Kafue, Northern Rhodesia as brought out by Phillips (10).

. . . it is not possible to say how close the post-Katanga granites approach the copperbelt under sedimentary cover, but there is evidence of some high temperature metamorphic effects, and the great post-Kundelungu gabbroid intrusions, which lie within the copperbelt, suggest the presence of underlying magma. . . .

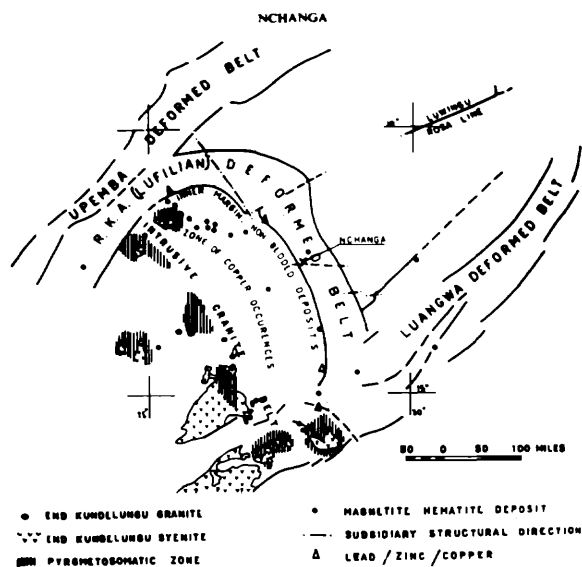


FIG. 4. The Northern Rhodesia-Katanga-Angola base metal belt. The R. K. A. Deformed Belt contains bedded copper-cobalt deposits. The Inner Margin deposits contain one or more of the following metals: lead, uranium, copper, zinc, cadmium, germanium, vanadium. Reproduced from Figure 89, page 269 of the volume "The Geology of the Northern Rhodesian Copper Belt" in the section on Nchanga written by D. M. McKinnon and N. J. Smit. The position of Nchanga is shown in the Deformed Belt, close to the Inner Margin.

On permeability (page 273) it is doubtful that the permeability of the Arkose has ever been so low as to prevent solutions passing both along and across the bedding . . . once having reached the Transition Beds, which probably have been more or less brecciated from the time of the formation of the syncline, solutions could find a ready avenue of movement.

On Genesis (page 274) the available evidence does not demonstrate beyond doubt the epigenetic or syngenetic origin . . . (but) the syngenetic theory is inadequate to explain the copper concentration in a variety of host rocks, and why only one sixth of the copper occurs in the formation most favourable for syngenetic copper, the Lower Banded Shales.

J. Jordaan, Chapter 10, Nkana.

Paragenesis . . . (page 319) Pyrite was the first mineral to form. Cattierite was next and continued until after the formation of carrollite. Carrollite continued to form during the early stages of deposition of chalcopyrite . . . which continued until after the formation of bornite. Chalcocite commenced to form after the deposition of the bornite, which it replaces . . . in general the greatest concentration of ore is in the Cherty ore and the least in the Low Grade argillite.

Mindola uranium deposit (page 325) occurred in the Transition zone between normal ore formation and the dolomites of the barren gap. The uranium occurs where copper mineralization is scarce, and the uranium grade varies in an inverse ratio to the copper content.

The age of the uranium mineralization was determined by Bowie (4) as about 520 million years. . . .

In the South ore body, the copper mineralization is limited to the highly contorted, fissile, and veined portions of the carbonaceous shales . . . no signs of metamorphic migration or hydrothermal alteration have been observed in the ores . . . the cherty ore which is heavily mineralized in sulphides, is similar in chemical and mineralogical composition to the Low Grade and Hanging Wall argillites, which are generally poorly mineralized. A physical control is indicated—the Cherty ore is laminated whereas the other beds are generally dense and massive.

The reader should not fail to notice that Mr. Jordaan describes a paragenetic sequence, and then goes on to say that there is no sign of metamorphic migration. It is only by metamorphic migration that the syngeneticists explain the paragenetic sequence of the growth and replacement of the ore minerals, one by the other.

O. Winfield, Chapter X, Chibuluma.

Economic mineralization (page 334) is confined to a sericitic quartzite, lying approximately 30 feet below the Upper Roan-Lower Roan contact . . . where it is well mineralized with chalcopyrite. Certain layers of the Ore quartzite rich in chalcopyrite or linnaeite, show megascopically conspicuous tourmaline as short prismatic crystals.

The correlation (page 339) between the rich chalcopyrite mineralization and the micaceous facies of the Ore formation is interpreted as being due to an originally greater argillaceous content in the center of the basin.

The chalcopyrite zones are arranged symmetrically on either side of the central pyritic zone and the sequence barren-chalcopyrite-pyrite, follows the direction of the aqueous currents as deduced from cross bedding . . . with selective precipitation of the copper nearer the shore line, where streams introduced copper, cobalt and iron in solution into a body of standing water . . . the streams must have had greater concentrations of metals than in the case of the more slowly deposited Roan Ore Shale type of mineralization. The aeolian sediments are completely barren of sulphides only a few feet below aqueous sediments containing ore.

Yes, Mr. Winfield believes in syngeneses.

J. B. Lee-Potter, Chapter XI, Baluba.

Chalcopyrite (page 350) is the main ore mineral. It is seen to enclose gangue minerals. The orebody lies in light green, tremolitic dolomite or schist, immediately overlying the Footwall conglomerate. Above are pyritic shales.

H. A. Groen, Chapter XII, Mufulira.

The ore zone is in about the middle of the Lower Roan, mostly in the A, B, and C horizons. Slump structures (page 428) occur in the B and C orebodies, espe-

cially near the fringes . . . banded quartzites in which mineralized and barren layers alternate can be traced laterally into breccias in which fragments of the mineralized layers carry the same mineralization as in the undisturbed rock. This indicates that the mineralization preceded the slumping.

These slump structures are one of Dr. Garlick's "proofs" of pre-folding mineralization. This critic might comment that he has seen many examples of mineralized (and unmineralized) breccia fragments and matrix in "breccia pipes" of obvious igneous origin.

In well bedded and cross bedded rocks, sulphides are usually concentrated on bedding planes . . . in B ore body bornite is abundant and in C, the dominant sulphide mineral is chalcocite. The sulphide minerals rarely exhibit crystal form, and commonly have the appearance of replacing both matrix and grains of the host rock, indiscriminately.

P. J. Pienaar, Chapter XIII, Bwana Mkubwa.

Most of the ore minerals are confined to the lower section of the ore formation which contains three major ore bodies separated by variable thicknesses of barren or poorly mineralized rocks . . . there is no evidence that concentration of ore minerals is preferential in certain rock types . . . although the ore at Bwana Mkubwa is entirely of supergene origin, the ore beds have been faulted and intruded by quartz gabbro, which is the ideal setting for the introduction of copper minerals. The suite of trace elements in the ore bodies is typical of many hydrothermal deposits, and their presence is difficult to explain, if the copper was syngenetically deposited.

Summary: for an igneous hydrothermal origin; Brock, McKinnon and Smit, and probably Jordaan and Piennar. for syngenetic origin; Mendelsohn, Garlick and Winfield, doubtful, or noncommittal; Schweltnus, Lee-Potter, and Groen.

Thus the vote may slightly favor the syngeneticists, but it is by no means a one sided decision.

Point 3.—There was an ample supply of copper from an eroding land-mass.

The syngeneticists, as exemplified by Dr. Brummer at the Symposium in New York, wave a hand in the direction of a Precambrian shore line, and point out that an eroding mass of crystalline rock, with an average amount of copper can yield a great deal of metal. Garlick (page 152) cites an average analysis of 329 American igneous rocks containing 0.01167%. Simple arithmetic shows that a cubic mile of such a rock contains 1.4 million tons of copper.

If the syngeneticists are right, and the copper was deposited in basins within then flat-lying marine sediments that have since been intensely folded, and much of them eroded, we must make an estimate of the amount originally deposited. What remains can only be a small percent of the original, since only a few infolded synclines remain. C. F. Davidson (5) guessed the original amount at 500 million tons of copper. That figure may be high, but probably not unreasonably so.

If the arc of coast line, along which these sediments were deposited was 400 miles long, and a total of 500 million tons of copper was deposited, then

along each of the 400 miles was a part—say 1,250,000 tons. This copper was concentrated in a relatively thin layer of sediments, in most cases less than 50 feet, yet all of that copper came out of 0.92 of a cubic mile of eroding rock.

How did all of the other detrital material get carried away beyond the basin in which the copper was deposited in such a thin layer?

The syngeneticists seem not to have tried to explain it. They say simply, “there was enough copper to account for all we see in these deposits.” Yes, there was copper, but also there was iron and manganese in greater quantity—where are they? There were lead, zinc, nickel and many other metals, if the source was an average crystalline rock with a few parts per million of most of the common metals.

The problem, therefore, is not so much quantity of copper, but concentrating all of it in one very narrow basin, at the same time getting rid of an enormous mass of detritus, which had to be carried farther away from shore. Also, there had to be a very selective precipitation of copper, cobalt, and only a little iron out of solutions which must have carried many other cations, if one depends as a source, the average composition of the typical crystalline rock.

Point 4.—This is the thesis that there is no question in any one’s mind, that the mineral deposition preceded the intense folding.

The syngeneticists have to adopt that view, for if the ore deposition were post-folding, they would not have a theory. Even if the ore were deposited before the folding, it still could have been introduced from an igneous source. However, the reasonable probability is that the igneous intrusion, and the intense folding were more or less contemporaneous. Such intense folding as took place in this area, has seldom, if ever in geological history, occurred without accompanying igneous intrusion.

Since the pre-folding date of deposition is such an essential feature of the theory, one would expect to have the evidence for it tabulated, and well documented so that the reader would be ready to agree that the proponents at least had made the best case possible. One would expect to find descriptions of ore beds displaced by post mineral faulting; of stretched deformable minerals such as chalcocite, and crushed brittle minerals such as pyrite, carrollite or linnaeite.

As a matter of fact, there has been remarkable little faulting on the Copperbelt. Figure 184 on page 471 shows a geological section in the Shovel Pit at the Bwana Mkubwa, and shows a fault which has brought Ore formation against the gabbro, but there is no indication that there was any actual ore displaced. Small faults are described at Nkana on page 308, but there seems to have been no displacement. Also at Nkana, displacement of the Ore formation by faulting at Mindola is described on page 327, but the statement is made:

Although the beds are displaced by a zone of faulting, the assay hangingwall and footwall of the orebody are undisturbed.

This is about as positive a statement as could be made, that the ore was deposited after the faulting!

Under the heading "Summary of the Evidence" written by Dr. Mendelsohn, p. 130–140, there is no mention whatsoever of faulting. One can be seasonably sure if ore were displaced by post mineral faulting, a big point would be made of it.

There is no mention of stretching out of the ore minerals under the intense folding, although thickening of the ore in "keels" of major folds is shown in many diagrams, as for instance in the Roan Antelope, Figures 131–132, at Nkana, Figure 104 and in Figure 3 accompanying the article by Darnley. Since no writer makes a point of this, the thickening must mean simple replacement of favorable beds which themselves were stretched and thickened during the folding.

Mendelsohn stated on page 141:

Within each deposit there is no direct association between folds and ore shoots. The ore shoots pass impartially, apart from attenuation and thickening, over folds and to considerable depths. Locally, ore shoots or mineral zones parallel folds, but this is exceptional.

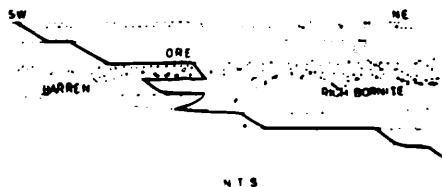


FIG. 5. Mufilira. Cross-bedded pebbly arkose with barren shale fragments, scoured into barren sediments in west, but overlying shale rich in disseminated bornite in ore body. The western margin of the orebody is indicated by a heavy line. Reproduced from Figure 36, page 137 in the volume "Geology of the Northern Rhodesian Copper Belt" in the section on Ore Genesis by F. Mendelsohn.

At Chibuluma, Winfield, a strong syngeneticist, makes some interesting comments:

Pebbles up to 4 inches, and rarely 8 inches in diameter in this sericitic grit are generally well rounded with one or two flattened surfaces. They consist of quartz and quartzite and show a concentration of coarse sulphide on their surfaces, especially on the pressure shadows, and commonly a smearing of the mica on the parts of the pebbles bearing the brunt of the pressure. The dissemination of sulphide rarely persists more than $\frac{1}{2}$ inch into the pebble so that the center is free of sulphide except along cracks . . . page 336. Pyrite occurs as small cubes replacing adjacent quartz and feldspar grains . . . and also tends to form rich, rounded clots up to 3 inches in diameter suggestive of diagenetic concretions . . . chalcopyrite in places follows shears across the bedding. Concentrations of coarser grains of chalcopyrite follow cross bedding laminations and are richer on bottomsets.

The sharp cut-off from barren to rich ore is shown in Figure section 36, page 137, shown here as Figure 5. How this is used to prove syngenetic deposition is not clear.

Slump breccias at Muliashi, Roan Antelope, are pictured in Figure 37, page 137, shown here as Figure 6. These breccia fragments contain pyrite, not copper ore, which may or may not be significant. A picture of such a

slump structure at Mufulira is shown on page 163. Dr. Garlick is quoted as saying that these features prove that mineralization preceded the slumping. Again it is not clear whether the mineralization is only pyrite, or copper minerals.

This slumping may be one of the best points to prove deposition of ore minerals during sedimentation. As no one of the epigeneticists has mentioned these features, the argument may have to be accepted. The writer has seen many fragments in breccias, replaced after brecciation.

The best evidence for mineral deposition after folding is that advanced by Brock (my Fig. 2). In Figure 7, these lines of alignment are shown on the Geologic Map. When one realizes that these ruler-straight lines connect the major mines into two groups, one on each side of the major anticline, through the most contorted kind of structures, it seems incredible to believe that primary deposition in the sediments as they lay flat on the sea floor, could be confined into basins, so that after folding the deposits are all perfectly aligned, and *are no where else*.

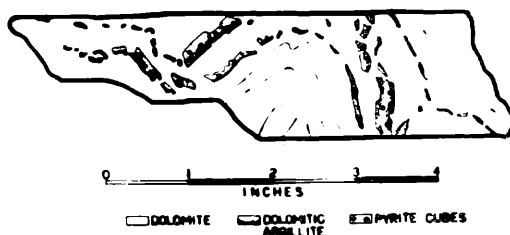


FIG. 6. Roan Antelope (Muliashi) Pre-consolidation dragfolding and brecciation of interbedded dolomite and pyritic dolomitic argillite. Reproduced from Figure 37, page 137 of the volume "The Geology of the Northern Rhodesian Copper Belt" in the section on Ore Genesis by F. Mendelsohn.

Beds of the Lower Roan carry ore only when the limb of a close fold crosses one of Brock's lines of alignment. These lines must be deep seated, and if not faults, at least channelways.

Point 5.—Possible introduction of material during metamorphism. Just as in the case of the date of deposition of the sulfide minerals as being pre- or post-folding, the syngenetic theory had to demand that the mineralization was pre-folding—so it is with the metamorphism and recrystallization that accompanied it. The syngenetic theory has to demand that the recrystallization was simply a remobilization of elements already present and that there was no introduction of material. To have admitted such an introduction, would have been to admit that solutions moved through the "impermeable rocks," whereas the non-existence of identifiable channelways is one of their major arguments against the hydrothermal theory.

It is a little amusing that the syngeneticists think that there is nothing strange in describing paragenesis of the sulfides, and stating that one was later than the other. This movement and replacement of one sulfide by another required that iron, and in some places iron and copper, had to be

phosphorus, and carbon dioxide . . . (and) there were adequate amounts of all the necessary elements present to form these minerals . . . and the growth of new minerals could have been achieved without the addition of material from an external source.

In Appendix 1, p. 487–493, analyses of rocks and minerals are tabulated. In the unmineralized rocks of the Katanga system, CO_2 varies from 0.15 to 43.35 percent (the last being a dolomite); there are two analyses for fluorine,

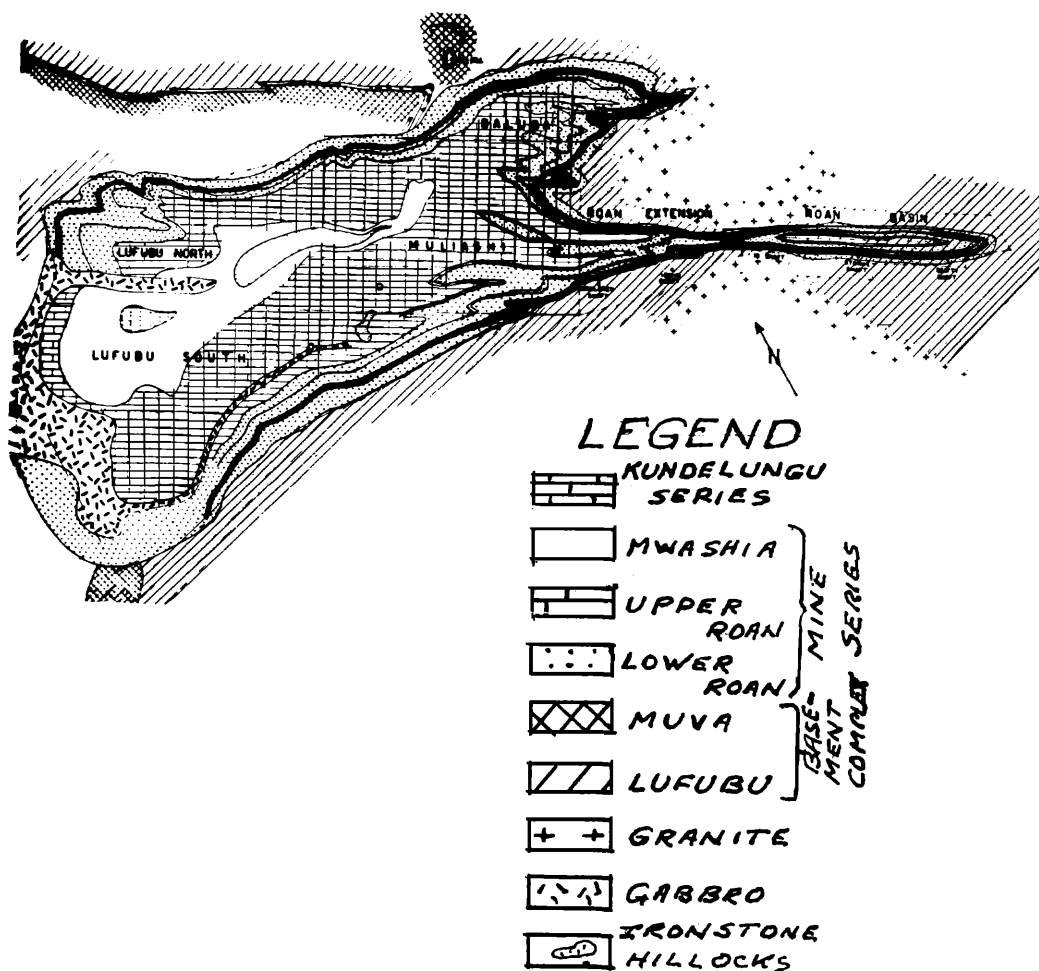


FIG. 8. Geological map of Roan-Muliashi basin. Ore formation in black. Reproduced from Figure 118, opposite page 350, of the volume "The Geology of the Northern Rhodesian Copper Belt," in the section on the Roan-Muliashi Basin by J. B. Lee-Potter. This map shows the remarkable persistence of the ore in an horizon of the Lower Roan formation which is on the flanks of a syncline. The southeast section of the mine lies on the flanks of a very tight fold, but the ore persists three quarters of the way around this limb.

one shows a trace, and the other 0.11% ; B_2O_3 is shown in one analysis only, as 0.03% ; and there are five analyses showing P_2O_5 which varies from .06 to .26 percent.

In rocks of the ore formations, CO_2 varies from a trace to 22.80% ; there is one analysis for fluorine, showing 0.09% ; there is one analysis for boron, showing 0.03% B_2O_3 ; and there are eight analyses for P_2O_5 , showing .03 to .26 percent.

These analyses are not very good support for Dr. Mendelsohn's "significant" amounts of the volatiles. All studies of ore deposits have indicated that probably a vast amount of mineralizers passed on through during metasomatism, leaving little trace ; but in Dr. Mendelsohn's thesis they could not pass on through, they had to stay where they are.

Meanwhile W. G. Darnley (6) in England and J. F. Vaes in Katanga were making detailed studies of the rocks of the Copperbelt, and both came to the same conclusion ; namely that there had been a great deal of introduction of material, particularly of potash. Dr. Darnley presented his paper at the January and February, 1960, meetings of the Institution of Mining and Metallurgy, and Dr. Vaes took part in the discussion at a later meeting of that organization. Hence, this information was available at the time "The Geology of the Northern Rhodesian Copperbelt" went to press in 1961, and deserved more attention than the six lines it received on page 131.

In summary, Dr. Darnley concluded that there was a progression of large scale metasomatism, which took place in the following stages :

1. Introduction of potash and boron, with formation of potash feldspar and sericite, enlargement of original tourmalines and formation of new grains of that mineral. Some of the rocks carry 10% K_2O and may become an "ore" of potash for fertilizer manufacture.
2. Introduction of quartz.
3. Introduction of ions which formed hematite, apatite, rutile, zircon and monazite, and the copper mineralization was about contemporaneous.
4. Albitization and scapolitization, probably connected with the large scale basic intrusions.

It is unnecessary to remind students of petrology and of economic geology, that large scale sericitization and albitization has gone on around many intrusions, involving the movement of millions of cubic yards of alkalis, and leaving little or no evidence of any channelways or openings.

Dr. Darnley is satisfied, however, that the copper mineralization was pre-folding, and he relates it to some volcanic activity, one example of which is a peculiar albite-biotite "sill" found in drill hole numbered NE 6B at Ndola. Ndola dome is just north of the Bwana Mkubwa mine.

Dr. Darnley also found evidence of enrichment in the ore of molybdenum and of silver.

Apatite, rutile and tourmaline are, of course, common accessories in many ore deposits of definite igneous hydrothermal origin. Tourmaline is a common constituent of high temperature vein deposits.

This writer (8) described zircon as a contact metamorphic mineral in Idaho.

In his discussion of Dr. Darnley's paper, Dr. Vaes expressed complete agreement with his conclusions that there had been large scale introduction of material during the metamorphism.

Point 6.—Evidence from Katanga. The volume "The Geology of the Northern Rhodesia Copperbelt" was written, as the name implies, to describe the deposits in Northern Rhodesia, and not the whole belt which continues northwestward into the Congo. Fortunately we have a fairly recent description of a part of that belt, by Oosterbosch (9). Dr. Schuiling had been expected to come to the United States in the Spring of 1960 to give a series of lectures, but he was prevented from doing so by ill health. It is to be hoped that he can come at a later time.

The copper and cobalt minerals of Katanga occur in two dolomitic formations which correspond in age with what in Northern Rhodesia is called the "Upper Roan" and hence the deposits are stratigraphically younger and higher than those south of the international border. The enclosing rocks are called the Mine Series. These are siliceous dolomites, well stratified, and form a tectonically competent group. The group appears between a complex of more or less arenaceous dolomite, much contorted, called "argillo-talcose rocks" (R.A.T.) and an upper series, called "Upper Sandstone Rocks" (R.G.S.).

The dolomites show no sign of derivation from limestones. All the carbonates are exclusively dolomites, and are in all probability of direct sedimentary origin. At the Fungurume district, described by Oosterbosch, the copper minerals are chalcocite and bornite; chalcopyrite and pyrite are practically absent.

In conclusion, Dr. Oosterbosch wrote:

It may be reasoned that: (a) the absence of mineralization in the breccia of argillo-talcose rock, which appears favorable (for deposition); the uniformity of mineralization which seem to indicate a horizontal rather than a vertical deposition of ore, and, to a less extent, the non-discovery of access channels, seem to us to be good reasons to consider the introduction of mineralization as subsequent to the breaking up the mine series into foliations, and

(b) In favor of the epigenetic hypothesis, mineralized formations are favorable formations (for such deposition) and the structure and texture of these layers suggest that there were in fact, at the beginning of deformation, the most favorable of all of the beds in the lower part of the dolomitic shale series.

In other words, the solutions were introduced after folding and brecciation started.

Thus, it is the plea of this reviewer that students of economic geology who review "The Geology of the Northern Rhodesian Copperbelt," read not only what Messers Mendelsohn and Garlick have written, but with an equally open mind, study the sections written by Brock, McKinnon and Smit, Jordaan and Pienaar, and also the writings of Dr. Darnley (6), and especially the pages of discussion (5, 7) that followed the presentation of that paper; and also read what is available on the deposits in Katanga.

Otherwise, one's conclusion may be that of the blind man described by McKinnon and Smit who felt only of the tail of the elephant; or, following another old proverb: "Be careful of half truths, since you may have hold of the wrong half."

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